SOFTWARE TEST LOG

SWE 3513 - Artificial Intelligence | Individual Assignment 2
Musanze Harvest Intelligence Service

Student	INEZA Benitha	Registration	25/27123
Email	ug2527123@ines.ac.rw	Verification	AI-A2-7123
Project path	E:\AI_A2_2527123	Seed	7123

1. Test environment and scope

Environment: Windows PowerShell; project virtual environment (.venv), Python, pytest and local FastAPI/Uvicorn server at 127.0.0.1:8000. Exact Python and package versions were not supplied. This record is compiled from actual user-reported terminal and browser outputs; original execution dates and times were not recorded.

2. Executed checks

ID	Test / command or route	Observed result	Outcome
T01	pytest -q	8 passed, 1 warning in 27.52s	PASS
T02	run_all.py --student-id ug2527123@ines.ac.rw --seed 7123	Completed; 273 rows; seed 7123; code AI-A2-7123	PASS
T03	Dataset SHA-256 from pipeline	759d98ee...cb3e014 (full hash below)	PASS
T04	predict.py --json samples/valid.json	Yield 12.2258 tons; risk 0; probability 0.2867; cluster 0	PASS
T05	predict.py --json samples/invalid.json	Missing required field: cooperative_support	PASS*
T06	Start Uvicorn server	Application startup complete	PASS
T07	GET /health	status: ok; verification AI-A2-7123	PASS
T08	GET /model-info	Correct student, 7 input features, seed 7123, model names and K=2	PASS
T09	GET /docs	Swagger UI opened; routes visible	PASS
T10	POST /predict with samples/valid.json	Valid prediction returned with verification code	PASS
T11	POST /predict omitting cooperative_support	HTTP 422 returned	PASS

* CLI validation outcome: A validation error was observed, but the invalid command exit code was not captured. The successful API POST response status code was not separately printed; a valid response body was observed.

3. Reproducibility and selected models

Full dataset fingerprint: 759d98eee29266a8d54d7de3c50c4a24c4ea50d4d5b38bf7082ed4ba3cb3e014

Selected regression: linear_regression; classification: logistic_regression; clustering: K = 2. The pipeline output records these selections. A second identical-fingerprint pipeline run has not been independently documented in the supplied logs.

4. Warning and outstanding verification

Pytest reported one StarletteDeprecationWarning concerning httpx/Starlette TestClient compatibility. It did not fail the test suite. Physical-device testing was not part of the provided evidence. Verify the CLI invalid-input exit code, exact software versions, technical report content, declaration signature and demonstration video separately before submission.

5. Recording integrity

This document records observed outcomes only; it is not a fabricated screenshot log or proof of unperformed checks.